5.2.6 Selection of drive

The choice of an electric drive depends on many factors. Some of the key factors are given below:

i. Depending on the nature cycle of the speed, torque characteristics, speed control, speed limit, performance and obligation, the level of operation, speed of performance.
ii. Uninterrupted operating requirements will depend on the acceleration and initial loss and performance change.
iii. The source and the ability to meet the source will consist of voltage, voltage optimization, power factor compatibility and other loads, and restoration of power adaptability
iv. Basic capital and running cost and maintenance is required.
v. Space and weight should be less.
vi. Environmental and area of the space be selected.
vii. Reliability.

i. Individual drive

As shown in figure 5.6, this drive is driven by an electric machine with an individual machine. It provides movement to various areas and all the operations are done by the individual machine.

A separate machine will change the direction and size. It is applicable in automobile systems, molecular machines, computers, radios, and television.

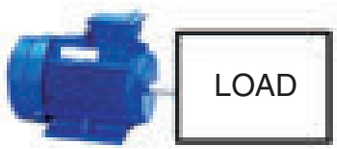


Fig. 5.6 Individual electric drive

ii. Group drive

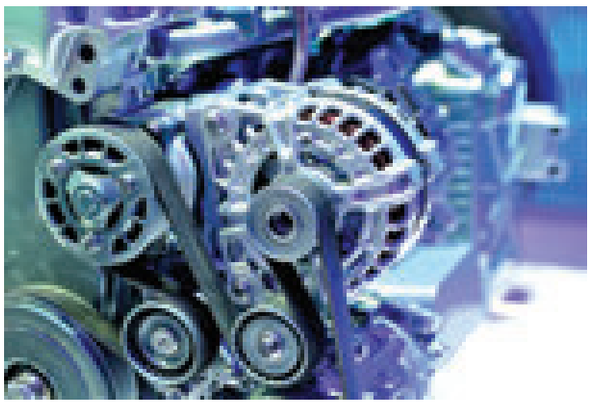


Fig. 5.7 Group electric drive

This drive has an electric motor. It is connected with one or more roller bearing and axis as shown in figure 5.7. This axis is fitted with belts or gears. With this, a group of electrical machines are operated.

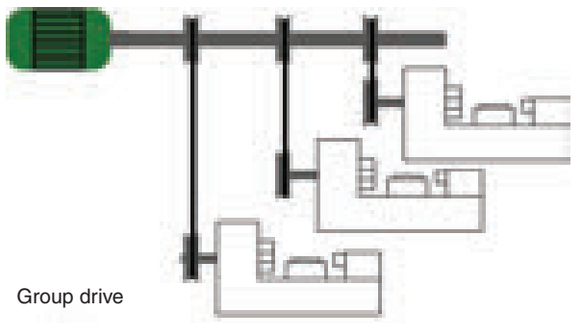


Fig. 5.8 Block diagram of group drive

a. Advantages

i. A large size of motor can be used insted of using small number of motors
ii. Low price.

b. Disadvantages

i. If the motor is in fault, the entire operation will be stopped
ii. The efficiency will be reduced due to losses by conduction of energy.

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